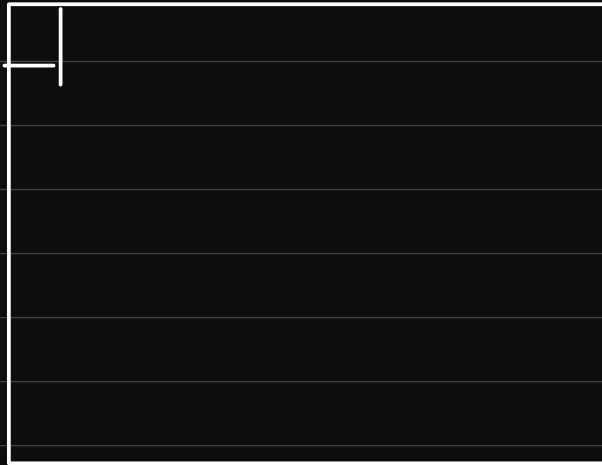
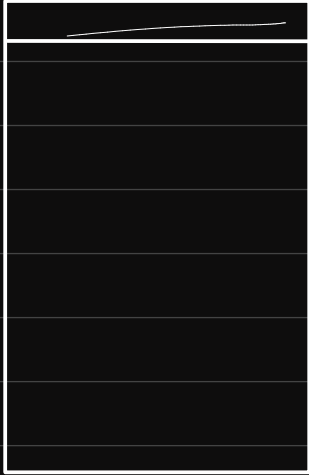




1653



943

$$L = (\gamma - p_i \cdot q_i)^2 + \lambda \sum (|p_i|^2 + |q_i|^2)$$

$$p_u = p_u - \alpha \frac{dL}{dp_u}$$

$$\frac{dL}{dp_u} = 2(\gamma - p_u \cdot q_i) \times q_i + \lambda 2p_u //$$

$$p_{u \sim} = p_{u_0} - \alpha (2q_i + \lambda 2p_u)$$

$$P_{un} = P_{u0} - \alpha (2(q_i^0 + \lambda P_u))$$

$$P_{un} = P_{u0} - 2\alpha (q_i + \lambda P_u)$$

$$P_u = P_{u0} - 2'(q_i + \lambda P_u) //$$

$$q_i^0 = q_i^0 - 2'(P_u + \lambda q_i^0) //$$